

Assemble a Valve SOP (1x Episode: 1 Valve)

One episode builds one valve. The table begins with the **valve body** clamped in its **holding fixture** on the assembly spot, just right of the center of the table. The body sits with its two **ports** pointing left and right, its open **neck** standing up on top, and its **shoulder** carrying two painted marks: the **CLOSED mark** at the front and the **OPEN mark** at the right. The neck starts empty. One **stem** and one **seal ring** are staged in the **start zone**; one **bonnet**, one **handle**, one **set screw**, one **set screw driver**, and one **tag** are staged in the right supply zone.

The order never changes: drop the stem into the bore, lay the seal ring in on top of it and press it home, thread the bonnet down hand tight, drop the handle onto the stem, drive the set screw to one click, cycle the handle open and back closed, then hang the tag. The seal ring does not go in until the stem is down on its shelf. The bonnet does not go on until the seal ring is seated all the way round. The handle is never cycled until the set screw has clicked. The tag goes on last, with the valve closed.

The right gripper does the work: it carries and drops the stem, lays and presses the seal ring, threads the bonnet, drops the handle on, holds the driver and drives the set screw, turns the handle through the cycle, and hangs the tag. The left gripper supports: it presses the seal ring down on the left side of the seat, squares the bonnet level before it goes on, holds the bonnet still while the handle goes on and while the handle is cycled, and holds the handle still while the set screw is driven. In Config L and M it also takes the stem and the seal ring from the start zone and hands each to the right gripper. The left gripper works the left side of the body only, and never reaches across the body.

The table is set up in one of three ways. Only the stem stand and the seal tray move; the body in its fixture, the bonnet stand, the handle stand, the screw tray, the driver cradle, and the tag tray are in the same place in all three.

- **Config L:** the stem stand and seal tray are at the back-left.
- **Config M:** the stem stand and seal tray are at the front-center, at the front edge, left of the tag hook.
- **Config R:** the stem stand and seal tray are at the front-right, left of the tag tray.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the stem stand and seal tray start at the back-left (**Config L**), the front-center (**Config M**) or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.

- **Same-side rule:** the gripper on the start zone's side takes the stem and the seal ring — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Hand-over rule:** in Config L and M the neck is the right gripper's work, so the left gripper **hands each part over** to the right gripper at the **hand-over spot**, above the middle of the table just left of the body: the left gripper holds the part still, the right gripper closes on it clear of the left gripper, and only then does the left gripper open and lift clear. The right gripper carries it on to the neck. Nothing is handed over in Config R.
- **Fixed roles:** the right gripper fits every part to the body, turns the bonnet, drives the set screw, cycles the handle, and hangs the tag; the left gripper presses, squares, and holds on the left side of the body. The order of Steps 1–7 is the same in all three configs.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the valve body in its fixture, the top of the neck, both marks on the shoulder, the stem stand and seal tray in the start zone for this episode's config, the right supply zone, and the tag hook on the front of the body.
3. The body is visible from above, so the open bore, the seal seat, and the top of the stem can be seen once the stem is in.
4. Both arms are at home with their grippers open.
5. The table is clear of everything except the fixture, the body, and the staged parts.
6. Both arms reach their own work without stretching: the right gripper reaches every station in the right supply zone, the top of the neck, the hand-over spot, and (Config R) the stem stand and seal tray; the left gripper reaches the left side of the bonnet, the left side of the seal seat, the handle lever, the hand-over spot, and (Config L and M) the stem stand and seal tray.

Materials checklist

1. One **valve body** is clamped in the **holding fixture** on the **assembly spot**, just right of the center of the table, with its two **ports** pointing left and right. It does not move when a gripper presses or turns on it.
2. The body's **bore** is empty, clean, and dry, and its **collar shelf** is clean.

3. The **neck** threads on the outside of the body are clean and undamaged, and the **seal seat** at the top of the neck is empty and clean.
4. The **CLOSED mark** is painted on the body **shoulder** at the front, and the **OPEN mark** on the shoulder at the right, a quarter turn apart.
5. One **stem** stands upright in the **stem stand**, with its **stem flat** and its **tapped hole** at the top end. The stem stand is in the **start zone** for this episode's config, and the other two start zones are bare:
 - **Config L**: back-left
 - **Config M**: front-center, at the front edge, left of the tag hook
 - **Config R**: front-right, left of the tag tray
6. One **seal ring** lies flat in the **seal tray**, right of the stem stand in every config, untwisted and unstretched, with no cracks, splits, or flat spots.
7. One **bonnet** sits on the **bonnet stand** at the right, threaded skirt down, with its **bonnet flats** upright and its thread clean.
8. One **handle** lies on the **handle stand** at the right, the right way up, with its **hub** bore facing down and its **pointer** end toward the front edge. Its **hub flat** and its **screw hole** are clean.
9. One **set screw** is staged head-up in the **screw tray** at the back-right, with clean threads and an unmarked socket.
10. One **set screw driver** stands in the **driver cradle** at the back-right, set to forward and to the validated torque setting.
11. One **tag** lies flat in the **tag tray** at the front-right, writing face up, with its **tag slot** open and toward the front edge. The **tag hook** on the front of the body is empty.
12. Keep the left side of the table clear apart from the stem stand and seal tray in Config L. The left gripper comes in from there onto the seal seat, the bonnet, and the handle.
13. Before collection, confirm by hand that the stem drops into the bore and lands on the collar shelf, that the seal ring drops into the seal seat around the stem without being stretched, that the bonnet threads down hand tight over the stem, that the handle drops onto the stem flat and sits down on the hub, that the set screw turns into the stem by hand, that the driver clicks at the set setting, and that the handle turns a quarter turn between the two marks.

Workspace layout

- **Assembly spot**: just right of the center of the table — the body stays clamped in its fixture here for the whole episode
- **Body shoulder**: the **CLOSED mark** at the front and the **OPEN mark** at the right
- **Front of the body**: the **tag hook**, empty
- **Start zone**: the stem stand with one stem and, right of it, the seal tray with one seal ring — back-left (**Config L**), front-center at the front edge, left of the tag hook (**Config M**), or front-right, left of the tag tray (**Config R**)

- **Hand-over spot:** above the middle of the table, just left of the body, where the stem and the seal ring change hands in Config L and M
- **Front-right:** the tag tray with one tag
- **Right supply zone:** the bonnet stand with one bonnet, and the handle stand with one handle
- **Back-right:** the screw tray with one set screw, and the driver cradle with the set screw driver
- **Left side:** kept clear apart from the start zone in Config L, so the left gripper can come in onto the seal seat, the bonnet, and the handle

Arm assignments

- **Left gripper:** in Config L and M, takes the stem and then the seal ring from the start zone and hands each over to the right gripper at the hand-over spot; presses the seal ring down on the left side of the seat while the right gripper presses the rest, squares the bonnet level over the neck before it is turned down, holds the bonnet still by its flats while the handle is dropped on and while the handle is cycled, and holds the handle lever still while the set screw is driven.
- **Right gripper:** takes the stem and the seal ring from their stand and tray in Config R, or from the left gripper at the hand-over spot in Config L and M; lowers the stem into the bore, lays the seal ring into the seat and presses it home, carries the bonnet and threads it down hand tight, drops the handle onto the stem, holds the set screw driver and drives the set screw, turns the handle open and closed, and hangs the tag on the hook.

Vocabulary

- **Valve body:** the part that everything else goes onto. It stays clamped in the holding fixture for the whole episode and is never lifted or turned.
- **Holding fixture:** the clamp on the assembly spot that holds the body still. It is never opened during an episode.
- **Ports:** the two round openings on the sides of the body. They point left and right and stay clear all episode.
- **Neck:** the round part standing up on top of the body. Its outside carries the thread the bonnet turns onto.
- **Bore:** the round hole down the middle of the neck that the stem drops into.
- **Collar shelf:** the ledge inside the bore that the stem's collar lands on. When the stem is down on the shelf it stops going down and stands at its own height.
- **Start zone:** where the stem stand and the seal tray are at the start of the episode — back-left (**Config L**), front-center at the front edge (**Config M**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.

- **Hand over:** the left gripper holds the stem or the seal ring still at the **hand-over spot** — above the middle of the table, just left of the body — the right gripper closes on it clear of the left gripper (the stem near its top end, the ring on the far side of its outside edge), and only then does the left gripper open and lift clear. Config L and M only, because the neck is the right gripper's work and the left gripper never works over the middle of the body.
- **Stem:** the shaft that drops into the bore and turns when the handle turns.
- **Collar:** the wider ring around the middle of the stem that lands on the collar shelf.
- **Stem flat:** the flat side milled on the top end of the stem. The handle's hub flat sits against it, so the handle turns the stem instead of spinning on it.
- **Tapped hole:** the threaded hole in the top face of the stem that the set screw goes into.
- **Seal seat:** the wide, flat-bottomed hollow at the top of the neck, around the stem, that the seal ring sits in.
- **Seal ring:** the rubber ring that drops into the seal seat around the stem and is squeezed by the bonnet.
- **Seated:** the seal ring lies flat in the bottom of the seal seat all the way round, with no part of it above the top of the neck and no part of it up on the neck's top edge.
- **Proud:** a part of the seal ring stands above the top of the neck instead of sitting down in the seat.
- **Twisted:** the seal ring is turned over on itself somewhere round the ring, so it does not lie flat in the seat.
- **Bonnet:** the threaded cap that turns down onto the neck and holds the seal ring and the stem in.
- **Bonnet flats:** the flat faces around the outside of the bonnet. They are the only place a gripper pinches the bonnet.
- **Drop:** the small fall a bonnet makes when it is turned counter-clockwise on the neck and the start of its thread lines up with the start of the body's thread. Feeling the drop before turning clockwise is how a bonnet is started straight.
- **Hand tight:** the bonnet has been turned clockwise until it stops turning, its bottom edge is down against the body's shoulder all the way round, and no gap shows under it. Nothing is turned past this point.
- **Shoulder:** the flat ledge around the base of the neck, below the bonnet. It carries the two marks and the bonnet lands on it.
- **CLOSED mark:** the painted mark on the shoulder at the front of the body. The handle points at it when the valve is closed.
- **OPEN mark:** the painted mark on the shoulder at the right of the body, a quarter turn from the CLOSED mark. The handle points at it when the valve is open.
- **Handle:** the lever that goes on the top of the stem and turns it.
- **Hub:** the round part of the handle that drops over the stem. Its bore has one flat side.

- **Hub flat:** the flat side inside the hub bore. It only goes on over the stem flat, so the handle drops on one way round.
- **Screw hole:** the hole straight down through the top of the hub that the set screw goes through into the stem's tapped hole.
- **Pointer:** the painted mark on the tip end of the handle. It is what is read against the CLOSED mark and the OPEN mark.
- **Lever:** the long arm of the handle, from the hub out to the pointer.
- **Sits down on the stem:** the bottom of the hub rests on the top of the bonnet, with no gap showing under the hub.
- **Set screw:** the short screw that drops through the handle's screw hole into the stem's tapped hole and clamps the handle onto the stem.
- **Set screw driver:** the hand driver in the cradle. Turned clockwise in the set screw, it clicks once when the screw reaches the set torque.
- **Driver vertical:** the bit points straight down and the driver body looks upright, not tilted, when seen against the handle hub.
- **Click:** the single sound and give of the driver when the screw reaches the set torque. It is the signal to stop turning.
- **Cycle:** turning the handle a quarter turn from the CLOSED mark to the OPEN mark and back again to the CLOSED mark.
- **Slips on the stem:** the handle turns but the stem below it does not, so the pointer moves past a mark or the handle spins freely on the stem.
- **Tag:** the flat card that hangs on the tag hook to identify the finished valve.
- **Tag slot:** the open cut in the top edge of the tag that slides sideways onto the tag hook.
- **Tag hook:** the small hook on the front of the body, below the shoulder, that the tag hangs on.
- **Hangs free:** the tag swings on the hook, does not rest on the table or the fixture, and does not cover either port.

Steps

Run Steps 1–7 in order on the one valve, then end the episode with Step 8. Steps 1 and 2 depend on where the stem stand and seal tray are: in Config L and M the **left gripper** takes the stem and then the seal ring from the start zone and **hands each over** to the right gripper at the hand-over spot; in Config R the **right gripper** takes them itself. Every other line, and Steps 3–8, are the same in all three configs.

Step 1: Seat the stem in the body

Goal: the stem stands down in the bore on its collar shelf, with its stem flat facing the front edge.

- Carry the stem in one go from the stem stand to the neck. Do not set it down on the body or the table on the way; in Config L and M it changes hands in the air at the hand-over spot.
- Keep the **left gripper** clear of the neck for the whole of this step.

1.1 Lift the stem

Look where the stem stand is before reaching for the stem.

- **IF the stem is at the back-left (Config L):** with the **left gripper**, pinch the stem below its **collar**, lift it straight up out of the stem stand, carry it upright forward and right to the **hand-over spot**, and hold it still. With the **right gripper**, close on the stem near its top end; then open the left gripper and lift it clear.
- **IF the stem is at the front-center (Config M):** with the **left gripper**, pinch the stem below its **collar**, lift it straight up out of the stem stand, carry it upright back to the **hand-over spot**, and hold it still. With the **right gripper**, close on the stem near its top end; then open the left gripper and lift it clear.
- **IF the stem is at the front-right (Config R):** with the **right gripper**, pinch the stem near its top end and lift it straight up out of the stem stand.

Then, in all three:

- With the **right gripper**, hold the stem upright above the neck.
- Turn the stem in the gripper until its **stem flat** faces the **front edge**.

1.2 Lower it into the bore

- With the **right gripper**, lower the stem straight down into the **bore** until its **collar** lands on the **collar shelf** and the stem stops going down.
- Lower it straight. Do not tilt it, wiggle it in, or push it down when it stops.
- If the stem catches on the way down, lift it straight back out with the **right gripper**, hold it upright again, and lower it once more.
- Open the **right gripper** only once the stem stands on its own in the bore.

Expected state: the stem stands upright in the bore, down on the collar shelf, with its stem flat toward the front edge, and the stem stand is empty.

Check: the stem stands straight up, it does not drop further when the gripper lets go, and its flat faces the front edge. If the flat has turned, pinch the stem top with the right gripper and turn it back until the flat faces the front edge.

Step 2: Seat the seal ring

Goal: the seal ring lies flat in the seal seat all the way round, with nothing proud and nothing twisted.

- **IF Config L or M:** with the **left gripper**, pinch the seal ring by its outside edge, lift it flat out of the seal tray, carry it level to the **hand-over spot**, and hold it still; the **right gripper** closes on the far side of its outside edge, then the left gripper opens and lifts clear. **IF Config R:** with the **right gripper**, pinch the seal ring by its outside edge and lift it flat out of the seal tray.
- With the **right gripper**, carry the ring level to above the neck and hold it there.
- Hold the ring by its outside edge only. Do not hook the gripper tip through the middle of it.
- With the **right gripper**, lower the ring straight down over the stem top and into the **seal seat**, and let it rest there.
- Lower the ring level. Do not stretch it, roll it, or pull it to make it fit.
- With the **left gripper**, press down on the ring on the **left side** of the seat and hold it there.
- With the **right gripper**, close the gripper and press the ring down into the seat with its tip: the back first, then the right side, then the front.
- Press the ring down only. Do not drag it round the seat and do not pry under it with the tip.
- Keep the **left gripper** on the left side until the rest of the ring is pressed home, then press the left side with the **right gripper** as the left gripper lifts clear.

Expected state: the seal ring lies flat in the seal seat all the way round, the seal tray is empty, the stem still stands down on its shelf, and the bonnet is still on its stand.

Check: the ring sits below the top of the neck all the way round, no part is **proud**, no part is up on the top edge of the neck, and no part is **twisted** over on itself. If a part stands proud, press it down again with the right gripper. If the ring is twisted, lift it out with the right gripper, lay it flat again, and press it home again.

Step 3: Thread the bonnet on

Goal: the bonnet is turned down hand tight on the neck, straight and down on the shoulder all the way round.

3.1 Bring the bonnet over the neck

- With the **right gripper**, pinch the bonnet by its **bonnet flats**, lift it off the bonnet stand, and hold it level above the neck, threaded skirt down.
- Carry the bonnet in one go. Do not set it down on the body or the table on the way.
- With the **left gripper**, touch the rim of the bonnet and square it level, so it sits flat over the neck and not tilted.
- With the **left gripper**, let go and move clear of the neck once the bonnet is level.
- With the **right gripper**, lower the bonnet straight down over the stem until its thread rests on the neck's thread.

3.2 Start the thread

- With the **right gripper**, turn the bonnet **counter-clockwise** slowly until it makes its small **drop** onto the start of the thread.
- With the **right gripper**, then turn the bonnet **clockwise** a quarter turn, open the gripper, take the flats again at the start of a fresh quarter turn, and turn again.
- If the bonnet binds, grinds, or stops turning in the first two turns, stop, lift it straight off with the **right gripper**, and start again from the **drop**.

3.3 Turn it down hand tight

- With the **right gripper**, keep turning the bonnet **clockwise** in quarter turns until it is **hand tight**: it stops turning and its bottom edge is down on the shoulder all the way round.
- Stop there. Do not force it further and do not use a tool on it.

Expected state: the bonnet is down on the shoulder all the way round, the bonnet stand is empty, both marks on the shoulder are still visible, and the top of the stem stands clear above the bonnet.

Check: no gap shows between the bonnet and the shoulder anywhere round the body, the bonnet sits square and not tilted, and the stem still stands straight up through it. If a gap shows on one side, turn the bonnet clockwise again with the right gripper until it is down all the way round. If it will not go down, end the episode at Step 8 and log the body and bonnet for a station check.

Step 4: Put the handle on the stem

Goal: the handle sits down on the stem with its hub flat on the stem flat, and its pointer over the CLOSED mark.

- With the **left gripper**, hold the bonnet still by its **bonnet flats** and keep holding it for the whole of this step.
- With the **right gripper**, pinch the handle by its **lever**, lift it off the handle stand, and hold it level above the stem with its **pointer** toward the **front edge**.
- With the **right gripper**, lower the handle straight down so the **hub** drops over the stem top and the **hub flat** goes onto the **stem flat**.
- Lower it straight. Do not force it down and do not twist the handle on the stem to make it drop.
- If the hub does not drop on, lift the handle straight up with the **right gripper**, turn it a little so the flats line up, and lower it again.
- With the **right gripper**, press the handle straight down until it **sits down on the stem**.
- With the **left gripper**, keep holding the bonnet until the handle is down.

Expected state: the handle sits down on the stem, its pointer is over the **CLOSED mark**, the handle stand is empty, and the set screw has not been touched.

Check: no gap shows under the hub, the pointer sits over the CLOSED mark, and the handle does not lift when the right gripper lets go. If the pointer is not over the CLOSED mark, lift the handle straight up with the right gripper, check that the stem flat still faces the front edge as in Step 1, and lower the handle on again.

Step 5: Drive the set screw

Goal: the set screw is driven down through the handle into the stem to one click, so the handle is clamped on the stem.

- With the **left gripper**, hold the **lever** still near the hub, and keep holding it for the whole of this step, so the handle cannot turn while the screw is driven.

5.1 Start the set screw

- With the **right gripper**, take the set screw from the screw tray by the sides of its head, threaded end down.
- Stand the screw in the handle's **screw hole**, straight and level, so it drops into the stem's **tapped hole**.
- With the **right gripper**, turn the screw **clockwise** a quarter turn, open the gripper, take the head again, and turn again, until the screw holds on its own when the gripper opens.
- If the screw goes in crooked, binds, or will not turn, stop turning, lift it back out with the **right gripper**, and set it in again. If it binds a second time, end the episode at Step 8 and log the stem for a station check.

5.2 Drive it to one click

- With the **right gripper**, lift the **set screw driver** out of its cradle by its handle and hold it **vertical** above the screw.
- With the **right gripper**, set the bit down into the screw head, keeping the driver **vertical**.
- Turn the driver **clockwise** in a steady move until it **clicks** once, then stop turning.
- If the bit slips out of the head, or the driver tilts off vertical, stop turning, lift it off, set it back down square, and turn again.
- With the **right gripper**, lift the driver straight up off the screw and return it to its cradle.
- With the **left gripper**, let go of the lever and move clear only once the driver is back in its cradle.

Expected state: the set screw is down in the handle hub, the driver has clicked once, the driver is back in its cradle, the screw tray is empty, and the pointer is still over the CLOSED mark.

Check: the screw sits down in the hub with the driver clicked once, the handle still sits down on the stem, and the pointer is still over the CLOSED mark. If the pointer has moved off the mark,

end the episode at Step 8 and log the handle for a station check.

Step 6: Cycle the valve open and closed

Goal: the handle is turned one full **cycle**, from the CLOSED mark to the OPEN mark and back, ending on the CLOSED mark.

- With the **left gripper**, hold the bonnet still by its **bonnet flats**, and keep holding it for the whole of this step, so the bonnet cannot turn with the handle.
- With the **right gripper**, pinch the **lever** near its **pointer** end.
- Turn the handle **clockwise** seen from above, in one steady sweep, until the **pointer** sits over the **OPEN mark**.
- Stop there. Do not turn the handle past the OPEN mark.
- Turn the handle **counter-clockwise** in one steady sweep, back until the **pointer** sits over the **CLOSED mark**.
- Open the **right gripper** and move it clear of the handle.
- With the **left gripper**, let go of the bonnet and move clear of the body.
- If the handle **slips on the stem** at any point, stop, end the episode at Step 8, and log the handle and set screw for a station check.

Expected state: the pointer is back over the CLOSED mark, the handle still sits down on the stem, the bonnet is still down on the shoulder all the way round, and both grippers are clear of the body.

Check: the pointer reached the OPEN mark and came back to the CLOSED mark, the handle did not slip on the stem, and the bonnet did not turn with the handle. If the bonnet has turned and a gap shows under it, end the episode at Step 8 and log the valve for a station check.

Step 7: Hang the tag

Goal: the tag hangs on the tag hook, writing facing out and upright, hanging free.

- Keep the **left gripper** clear of the body for the whole of this step.
- With the **right gripper**, pinch the tag by its bottom edge, lift it flat out of the tag tray, and hold it upright with its writing facing out and its **tag slot** at the top.
- Carry the tag in one go. Do not set it down on the body or the table on the way.
- With the **right gripper**, bring the tag to the **tag hook** on the front of the body and slide the **tag slot** sideways onto the hook.
- Open the **right gripper** and let the tag hang, then move the gripper clear of the body.

Expected state: the tag hangs on the tag hook, the tag tray is empty, and both grippers are clear of the valve.

Check: the tag is on the tag hook, its writing faces out and is upright, and it **hangs free** without resting on the table or covering a port. If the tag hangs crooked or rests on something, take it by its bottom edge with the right gripper, lift it off the hook, and hang it again. If the tag falls to the table, pick it up with the right gripper and hang it again.

Step 8: End the episode

Goal: recording ends with the valve built, closed, and tagged, and both arms home.

1. Confirm the stem is down in the bore, the seal ring is seated all the way round, the bonnet is hand tight on the shoulder, the handle sits down on the stem with the set screw clicked, the pointer is over the CLOSED mark, the tag hangs on the tag hook, every tray and stand is empty, and the set screw driver is back in its cradle.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. With recording off, set the driver to reverse, or use a hand driver, and back the set screw out of the handle. Return the driver to forward and to the validated torque setting.
2. Lift the handle straight up off the stem and lay it on the handle stand, the right way up, with its pointer toward the front edge.
3. Unthread the bonnet counter-clockwise off the neck and set it on the bonnet stand, threaded skirt down, flats upright.
4. Lift the seal ring out of the seal seat and lay it flat in the seal tray, untwisted.
5. Lift the stem straight up out of the bore and stand it in the stem stand. Set the stem stand and the seal tray, seal tray right of the stand, in the start zone for the next episode's config — back-left (Config L), front-center at the front edge (Config M), or front-right (Config R).
6. Take the tag off the tag hook and put it in the used-tag bin.
7. Wipe grit, dust, and rubber crumbs out of the bore, off the collar shelf, out of the seal seat, and off the neck thread. Wipe the bonnet thread and the stem.
8. Stage one set screw head-up in the screw tray, and leave the handle's screw hole and the stem's tapped hole empty.
9. Lay one fresh tag flat in the tag tray, writing face up, tag slot toward the front edge.
10. Inspect the parts. Replace the seal ring if it is cracked, split, flat-spotted, or stretched so far that it no longer lies flat in the seat. Replace the set screw if its threads are damaged or its socket is rounded. Replace the handle if its hub is cracked or its hub flat is worn round. Replace the stem if it is bent or its tapped hole no longer holds the screw. Replace the bonnet or the body if a thread binds or no longer pulls down hand tight. Tighten the holding fixture if the body moves under a gripper.

11. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R (the stem stand and seal tray at the front-right, with the other supplies). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the stem stand and seal tray are not in the start zone for the config; a gripper reaches across the table for the stem or the seal ring; in Config L or M the left gripper carries the stem or the ring to the neck itself, or the right gripper reaches to the left for one, instead of a hand-over at the hand-over spot; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the same-side rule, and the hand-over rule (the left gripper takes the stem and the seal ring from the start zone in Config L and M and hands each over to the right gripper at the hand-over spot; in Config R the right gripper takes them itself; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the stem stand is before the first reach, then follow that config's IF lines through Steps 1 and 2.

Violation: Stem mishandled on the way in

- **Visible cue:** the stem is set down on the body or the table on the way from its stand, is carried in more than one move, or is dragged across the neck.
- **SOP rule broken:** Step 1 (carry the stem in one go, straight from the stem stand to the neck).
- **Coaching note:** one lift, straight over the neck, straight down.

Violation: Stem not seated straight and down

- **Visible cue:** the stem goes in tilted, is wiggled or pushed down after it stops, or it is left standing high because the collar never lands on the collar shelf.
- **SOP rule broken:** Step 1.2 (lower the stem straight down until the collar lands on the collar shelf; if it catches, lift it straight out and lower it again).
- **Coaching note:** straight down until it stops on its own. If it catches, take it out, do not push.

Violation: Stem flat not facing the front edge

- **Visible cue:** the stem's flat faces a side or the back when the gripper lets go, and the episode goes on to the seal ring anyway.
- **SOP rule broken:** Step 1 (turn the stem so its flat faces the front edge, and turn it back with the right gripper if it has moved).
- **Coaching note:** set the flat to the front before you let go. The handle can only land on that flat.

Violation: Seal ring mishandled on the way in

- **Visible cue:** the ring is picked up through its middle with the gripper tip hooked in it, is set down on the neck or the table on the way, or is stretched, rolled, or pulled to make it drop into the seat.
- **SOP rule broken:** Step 2 (hold the ring by its outside edge, carry it level in one go, and lower it in without stretching, rolling, or pulling it).
- **Coaching note:** outside edge, level, straight down. It should drop in on its own.

Violation: Seal ring not seated all the way round

- **Visible cue:** part of the ring stands proud of the top of the neck, sits up on the top edge of the neck, or is twisted over on itself, and the step moves on.
- **SOP rule broken:** Step 2 (the ring lies flat in the seat all the way round, with nothing proud, nothing on the top edge, and nothing twisted).
- **Coaching note:** press all the way round and look all the way round before you reach for the bonnet.

Violation: Seal ring left side not held

- **Visible cue:** the left gripper is off the left side of the seat while the right gripper presses the rest of the ring, and the ring lifts, shifts round the seat, or pops out on the left.
- **SOP rule broken:** Step 2 (the left gripper presses the ring down on the left side and holds it until the rest is pressed home).
- **Coaching note:** left gripper pins the left side first, then press round with the right.

Violation: Bonnet started crooked

- **Visible cue:** the left gripper never squares the bonnet level, or the bonnet is turned clockwise without the counter-clockwise **drop** first, or it goes on visibly tilted and is turned anyway.
- **SOP rule broken:** Step 3.1–3.2 (square the bonnet level with the left gripper, then back it counter-clockwise to the drop before turning it clockwise).
- **Coaching note:** level it, feel the drop, then turn. A bonnet started crooked cuts the thread.

Violation: Binding bonnet forced

- **Visible cue:** the bonnet binds, grinds, or stops turning early and the gripper keeps turning it, instead of it being lifted off and started again from the drop.
- **SOP rule broken:** Step 3.2 (if the bonnet binds, grinds, or stops in the first two turns, lift it straight off and start again from the drop).
- **Coaching note:** stop at the first bind. Forcing it wrecks the neck thread.

Violation: Bonnet not turned down hand tight

- **Visible cue:** the episode goes on to the handle with a gap showing between the bonnet and the shoulder anywhere round the body, or with the bonnet sitting tilted.
- **SOP rule broken:** Step 3.3 (turn the bonnet clockwise until it stops and its bottom edge is down on the shoulder all the way round).
- **Coaching note:** look all the way round the joint for a gap before you pick up the handle.

Violation: Bonnet forced past hand tight

- **Visible cue:** the gripper keeps turning the bonnet after it has stopped, or a tool is brought onto the bonnet to turn it further.
- **SOP rule broken:** Step 3.3 (stop at hand tight; do not force it further and do not use a tool on it).
- **Coaching note:** hand tight is the finish line. Extra turning only crushes the seal.

Violation: Handle put on out of clock

- **Visible cue:** the handle is lowered with its pointer toward a side or the back, or it ends the step with its pointer somewhere other than over the CLOSED mark, and the episode goes on.

- **SOP rule broken:** Step 4 (hold the handle with its pointer toward the front edge, and end the step with the pointer over the CLOSED mark).
- **Coaching note:** pointer to the front, over the CLOSED mark, every time.

Violation: Handle not seated on the stem

- **Visible cue:** the handle is forced down or twisted on the stem to make the hub drop on, or it is left standing with a gap under the hub, or it lifts when the gripper opens.
- **SOP rule broken:** Step 4 (lower the handle straight down onto the flats without forcing or twisting it, and press it down until it sits down on the stem with no gap under the hub).
- **Coaching note:** line the flats up, drop it on, press it home. Never twist it down.

Violation: Bonnet not held while the handle goes on

- **Visible cue:** the left gripper is off the bonnet flats while the handle is lowered and pressed down, or it lets go before the handle is down, and the bonnet turns or lifts.
- **SOP rule broken:** Step 4 (the left gripper holds the bonnet by its flats for the whole of the step and lets go only once the handle is down).
- **Coaching note:** left gripper pins the bonnet, then the right gripper presses the handle home.

Violation: Set screw started crooked or forced

- **Visible cue:** the set screw goes in leaning, misses the stem's tapped hole, or binds, and the gripper keeps turning it instead of lifting it out and setting it in again.
- **SOP rule broken:** Step 5.1 (stand the screw straight in the screw hole, and if it goes in crooked, binds, or will not turn, lift it out and set it in again).
- **Coaching note:** stop at the first bind. A forced screw strips the stem.

Violation: Driver not held vertical

- **Visible cue:** the driver is tilted when the bit is set down, it tilts off upright while turning, or the bit slips out of the screw head and turning continues.
- **SOP rule broken:** Step 5.2 (hold the driver vertical, and if the bit slips or the driver tilts, stop, set the bit back down square, and turn again).
- **Coaching note:** square it, then turn. A tilted driver rounds the socket.

Violation: Set screw not driven to one click

- **Visible cue:** the driver is lifted off before it clicks, or it keeps turning after the click, or it is worked until it clicks a second time on the same screw.
- **SOP rule broken:** Step 5.2 (turn clockwise until the driver clicks once, then stop turning and lift it straight up).
- **Coaching note:** one click, hands off. The click is the finish line, not a checkpoint.

Violation: Handle not held while the set screw is driven

- **Visible cue:** the left gripper is off the lever while the screw is started or driven, or it lets go before the driver is back in its cradle, and the handle turns off the CLOSED mark.
- **SOP rule broken:** Step 5 (the left gripper holds the lever still for the whole of the step and lets go only once the driver is cradled).
- **Coaching note:** hold the lever, then drive. A handle that turns while you drive ends up off its mark.

Violation: Cycle not run to both marks

- **Visible cue:** the handle is never cycled, the pointer stops short of the OPEN mark, it is turned past the OPEN mark, it never comes back to the CLOSED mark, or the episode goes on with the pointer off the CLOSED mark.
- **SOP rule broken:** Step 6 (turn the handle clockwise to the OPEN mark, stop there, then counter-clockwise back to the CLOSED mark).
- **Coaching note:** mark to mark, one sweep each way, and finish on CLOSED.

Violation: Handle slipping on the stem ignored

- **Visible cue:** the handle turns while the stem below it does not, or the handle spins past a mark, and the episode carries on to the tag.
- **SOP rule broken:** Step 6 (if the handle slips on the stem, stop and end the episode at Step 8).
- **Coaching note:** a slipping handle means the set screw did not take. Stop the episode and flag the valve.

Violation: Bonnet not held during the cycle

- **Visible cue:** the left gripper is off the bonnet flats while the handle is turned, and the bonnet turns with the handle or lifts off the shoulder.
- **SOP rule broken:** Step 6 (the left gripper holds the bonnet by its flats for the whole of the cycle).
- **Coaching note:** pin the bonnet before you turn the handle, or you unthread what you just tightened.

Violation: Tag hung wrong

- **Visible cue:** the tag hangs with its writing facing the body, upside down, or crooked; it rests on the table, the fixture, or a port; it covers a port; or it is put on the handle, the bonnet, or a port instead of the tag hook.
- **SOP rule broken:** Step 7 (slide the tag slot onto the tag hook so the tag hangs free, writing facing out and upright).
- **Coaching note:** slot onto the hook, writing out, and let it swing clear.

Violation: Work done out of order

- **Visible cue:** the steps run out of sequence, for example the seal ring goes in before the stem is down, the bonnet goes on before the ring is seated, the handle is cycled before the set screw has clicked, or the tag goes on before the cycle is run.
- **SOP rule broken:** Steps 1–7 (stem, seal ring, bonnet, handle, set screw, cycle, tag, in that order).
- **Coaching note:** the order is the task. Say the next step's name before you move.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper carrying the stem, the seal ring, the bonnet, the handle, or the tag, turning the bonnet, driving the set screw, or turning the handle; or the right gripper holding the bonnet during Steps 4 and 6, or holding the lever during Step 5.
- **SOP rule broken:** Steps 1–7 (the right gripper carries and fits every part, turns the bonnet, drives the screw, cycles the handle, and hangs the tag; the left gripper presses the left side of the seal ring, squares the bonnet, holds the bonnet, and holds the lever).
- **Coaching note:** right gripper does the work, left gripper holds and steadies.

Violation: Part or supply knocked off its spot

- **Visible cue:** the stem, seal ring, bonnet, handle, set screw, driver, or tag is dropped on the body, the table, or the floor; a tray, stand, or cradle is tipped, pushed off its spot, or spilled; or the body is knocked out of square in its fixture.
- **SOP rule broken:** Steps 1–7 (keep the parts, the trays, the stands, the cradle, and the body on their spots through the assembly).
- **Coaching note:** work slower and lower over the table, and keep every carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the end state is confirmed, the driver is left out of its cradle, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 8 (confirm the end state with the driver cradled, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.

- An arm or gripper fails, drifts, or reports a motor error.
- The set screw driver does not click at the set setting, clicks early, or slips its setting.
- The seal ring is split, flat-spotted, or moulded out of shape, so a correctly laid ring will not lie flat in the seat.
- The neck or bonnet thread is galled or cross-cut before the episode, so a correctly started bonnet binds anyway.
- The stem is bent, or its tapped hole is stripped, so a correctly started set screw spins without clamping.
- The stem flat or the handle's hub flat is worn round, so a correctly fitted handle turns on the stem.
- The tag hook is bent or broken off, so a correctly hung tag will not stay on it.
- The holding fixture is loose, so the body turns however carefully a gripper works on it.

Annotation subtasks (from SOP)

1. Carry the stem from its stand and lower it into the bore
2. Hand the stem or the seal ring from the left gripper to the right gripper at the hand-over spot (Config L and M)
3. Turn the stem so its flat faces the front edge
4. Carry the seal ring from the tray and lay it into the seal seat
5. Press the seal ring home all the way round
6. Carry the bonnet from its stand and square it over the neck
7. Thread the bonnet down hand tight
8. Drop the handle onto the stem and press it down
9. Start the set screw by hand
10. Pick up the driver and drive the set screw to a click
11. Return the driver to its cradle
12. Turn the handle from the CLOSED mark to the OPEN mark and back
13. Hang the tag on the tag hook
14. Confirm the end state, return both arms home, and end the episode